

PhD Research Proposal — Working Draft v0.3

PhD Research Proposal — Working Draft v0.3

Status: WORKING DRAFT — NOT SUPERVISOR-APPROVED, NOT A SUBMISSION CANDIDATE.
Prepared by Ali Hamed for the supervisor meeting of Wednesday 2026-08-12, 09:00. Supervisors: Prof. Iris Reinhartz-Berger, Prof. Arnon.

This is the Word proposal document requested on 2026-08-05 (E15/A08-06). It is maintained separately from the tracking document, as instructed. v0.3 delivers **Chapter 3 in full** — the deliverable due at this meeting — and holds the remaining chapters at the status each has actually reached. Nothing is presented as further along than it is.

Title — decision required

The v0.2 title is "Reusable Human Judgment for Auditable, Reliable, and Transferable Agentic AI Assessment." Three of its words — *reusable*, *auditable*, *transferable* — are precisely the words the 2026-08-05 live edit removed from the research-question headline (E5: reuse belongs in a sub-question; E7: lean toward dropping "auditable"/"transferable"/"end-to-end"). The §3 wording now complies; **the title does not**. Leaving it is a visible inconsistency in a submitted proposal.

Three candidates, aligned to the current headline wording, for the supervisors to choose or replace:

#	Candidate title	Rationale
T1	<i>Human Judgment in Agentic-AI Variability Exploration: Capture, Governance, and Reliable Co-Reasoning</i>	Mirrors the U-RQ almost verbatim; carries "reliable" (retained per E7) and drops all three removed words
T2	<i>Reliable Human–AI Co-Reasoning for Variability Exploration in Guideline Operationalization</i>	Leads with the outcome; keeps the domain-neutral framing required by E13
T3	<i>From One-Off Correction to Governed Judgment: Human Expertise in Agentic AI Assessment</i>	Leads with the gap (§3.3) rather than the artefact; strongest "open question" framing per E4

Decision `D-TITLE-01` — owner: supervisors, at the 2026-08-12 meeting. Until it is logged, the v0.2 title stands unchanged in the record.

Chapter plan and current status

Structure per the confirmed chapter skeleton (E13, machine-derived record, pending confirmation):

#	Chapter	Status at 2026-08-12	Note
1	Introduction	Not started — deliberately	E13 places it last; written once the questions are signed off
2	Literature Survey	Not started — deliberately	E15: <i>think about it, do not start it</i> . Four structural options prepared; searches QL-01–QL-05 frozen and not run
3	Gap and Research Questions	COMPLETE — this deliverable	Full text below. Wording provisional pending A08-01 / D-RQ-01 / D-RQ-02
4	Research Methodology	Not started — deliberately	Design Science per Prof. Penina's course; per-RQ study/artifact/design/evaluation. Nine artifact options prepared, none selected
5	Preliminary Results	Partially available	Software/modelling only; mechanism and readiness evidence. Medical = infrastructure only
6	Plan	Not started	E14: ~3-month semester-aligned blocks over a 3-year horizon; never month-by-month

What "not started" means here: it is compliance with an instruction, not slippage. Iris asked for §2 and §4 to be *thought about* and not begun; the thinking is delivered as a separate options note.

Chapter 3 — Gap and Research Questions

The complete chapter follows as a separate document in this package

(02 - Chapter 3 - Gap and Research Questions), and is incorporated here by reference so the two

never drift. Summary of what it argues:

§3.1 frames the interpretive step: once a system detects that an artefact deviates from a norm, something must decide what the deviation *means* — legitimate alternative, error, domain convention, guideline defect, or genuine ambiguity. This is where automation is weakest and expertise most expensive.

§3.2 positions four bodies of work and states, for each, what it establishes and what it leaves open: automated model assessment (detects deviation, does not interpret it); LLM and agentic modelling assistance (documents *that* expert correction is needed, not *when* to request it or *what* to keep); human-in-the-loop and oversight research (supplies mechanisms and governance expectations, not a selection policy for agentic assessment under an attention budget); and variability engineering (formalises *designed* variability, not *observed interpretive* variability). It also defines both constructs

— "variability" in this thesis's sense, and "guideline operationalization" — against the established literature.

§3.3 states the gap: the design knowledge for connecting selective expert intervention, structured judgment capture, and scope-aware reuse does not exist in tested, generalizable form, and neither does an evaluation approach able to say which parts of such a lifecycle transfer. It argues why expert assessment judgment resists both mature capture paradigms — it is case-grounded, scoped, contestable, and authority-bearing — and it explicitly does **not** presuppose that reusable judgment improves assessment.

§3.4–§3.7 give the main question and SQ1–SQ3, each with an argument for why it is open, an evaluation criterion built into the question itself, and the wording decisions still outstanding.

§3.8–§3.11 compose the three questions into one programme, position VEGO-AI as motivating case rather than object of study, bound the scope, and state the evidence status plainly, including what a negative result would mean for each question.

Chapter 5 — Preliminary Results (status only)

Reported here as *status*, not as results, because the chapter is not written:

Element	State
Software/modelling offline evidence	Mechanism, architecture and readiness evidence from the offline replay series; no effectiveness claim
Independent expert labels (EXP-005)	0 of 24 generalization-safe labels supplied; ≥ 20 required for any quantitative claim
Medical instantiation	Infrastructure and feasibility only; 0 of 6 entry gates pass
Accuracy / generalization / clinical performance / effort reduction	No claim made or supported

Which preliminary results may appear in the proposal — and under what label — is an open question for the supervisors (see the thinking note, Part 3).

Standing evidence boundary

Unchanged by anything in this version:

- No accuracy, generalization, clinical-performance or effort-reduction claim appears anywhere in this proposal, and none may until the label and gate counts change.
- The literature searches are protocol-ready and **not executed**; no novelty or review-completeness statement is made.
- The research-question wording is **provisional** pending **A08-01** verification and logged

D-RQ-01/D-RQ-02 decisions.

- The 2026-08-05 meeting record is machine-derived with inferred speakers and is pending participant

confirmation; every reference to it carries its E-identifier so it can be checked in one step.

v0.3 supersedes v0.2 for Chapter 3 only; all other v0.2 content and the v0.1 baseline remain in the record. Version history and rationale are preserved in the decision and change log.